

Rank

EE263

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Rank of a matrix

we define the *rank* of $A \in \mathbb{R}^{m \times n}$ as

$$\text{rank}(A) = \dim \text{range}(A)$$

(nontrivial) **facts:**

- ▶ $\text{rank}(A) = \text{rank}(A^T)$
- ▶ $\text{rank}(A)$ is maximum number of independent columns (or rows) of A
hence $\text{rank}(A) \leq \min(m, n)$

Rank of a matrix

- ▶ **rank**(A) is maximum number of independent columns of A
 - ▶ to see this, notice that if the columns of A are independent, then the number of columns r is the rank, since the columns are a basis for the range
 - ▶ and if not, then there must be one column in the span of the others, so remove it, and repeat if necessary
 - ▶ all other independent sets of columns must have no more than r elements.
- ▶ proof of **rank**(A) = **rank**(A^T) uses QR, to come

Conservation of dimension

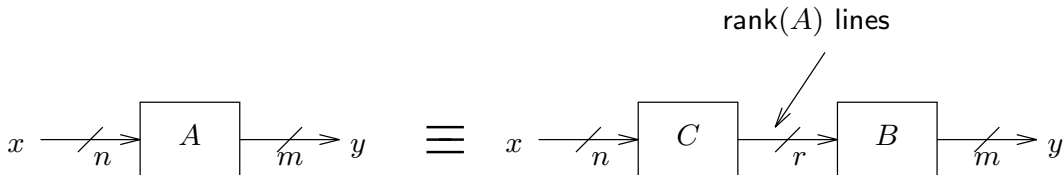
$$\mathbf{dim\ range}(A) + \mathbf{dim\ null}(A) = n$$

- ▶ **rank**(A) is dimension of set 'hit' by the mapping $y = Ax$
- ▶ **dim null**(A) is dimension of set of x 'crushed' to zero by $y = Ax$
- ▶ 'conservation of dimension': each dimension of input is either crushed to zero or ends up in output
- ▶ roughly speaking:
 - ▶ n is number of degrees of freedom in input x
 - ▶ **dim null**(A) is number of degrees of freedom lost in the mapping from x to $y = Ax$
 - ▶ **rank**(A) is number of degrees of freedom in output y
- ▶ proof using QR

Coding interpretation of rank

$$\text{rank}(BC) \leq \min\{\text{rank}(B), \text{rank}(C)\}$$

- ▶ hence if $A = BC$ with $B \in \mathbb{R}^{m \times r}$, $C \in \mathbb{R}^{r \times n}$, then $\text{rank}(A) \leq r$
- ▶ converse: if $\text{rank}(A) = r$ then $A \in \mathbb{R}^{m \times n}$ factors as $A = BC$ with $B \in \mathbb{R}^{m \times r}$, $C \in \mathbb{R}^{r \times n}$
- ▶ $\text{rank}(A) = r$ is minimum size vector needed to faithfully reconstruct y from x



Coding interpretation of rank

- ▶ $\text{rank}(BC) \leq \text{rank}(B)$ because $\text{range}(BC) \subset \text{range}(B)$
- ▶ transpose implies $\text{rank}(BC) \leq \text{rank}(C)$
- ▶ factorization converse comes from QR

Application: fast matrix-vector multiplication

- ▶ need to compute matrix-vector product $y = Ax$, $A \in \mathbb{R}^{m \times n}$
- ▶ A has known factorization $A = BC$, $B \in \mathbb{R}^{m \times r}$
- ▶ computing $y = Ax$ directly: mn operations
- ▶ computing $y = Ax$ as $y = B(Cx)$ (compute $z = Cx$ first, then $y = Bz$): $rn + mr = (m + n)r$ operations
- ▶ savings can be considerable if $r \ll \min\{m, n\}$

Full rank matrices

for $A \in \mathbb{R}^{m \times n}$ we always have $\text{rank}(A) \leq \min(m, n)$

we say A is *full rank* if $\text{rank}(A) = \min(m, n)$

- ▶ for **square** matrices, full rank means nonsingular
- ▶ for **skinny** matrices ($m \geq n$), full rank means columns are independent
- ▶ for **fat** matrices ($m \leq n$), full rank means rows are independent